

Weekly Report 7

Project title: Scot Forge

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Team members:

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Progress Report:

In our updated model, the optimized strategy reduced daily penalties by roughly 264 compared to actual operations, while monthly penalties remained about the same (~15). We are continuing to refine the model, particularly around monthly constraint handling.

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Forecasting 2022-2024 (2-day ahead, blended)...\nRunning 2-day-ahead optimization (horizon=60, v4)...\nEvaluating penalties...\nActual                                     | Daily:  420 | Monthly:  15\nOptimized (v4)                           | Daily:  156 | Monthly:  15
```

Tier wise comparison is given below.

A	B	C	D	E	
Actual_Daily_Violation	TRUE	Y			
	act				
	Column Labels				
		1	2	3	Grand Total
Count of Date		71	47	302	420
Opt_Daily_Violation	TRUE	Y			
	opt				
	Column Labels				
		1	2	3	Grand Total
Count of Date		22	25	109	156

Action Items:

- Improve Monthly penalties.
- Improve optimization by including financial aspect.